Month	Observation	Deviation from the 3% Target	Deviations below the Target	Squared Deviations below the Target
March	−1	−4	−4	16
April	−4	−7	−7	49
May	4	1	—	—
June	2	−1	−1	1
July	0	−3	−3	9
August	4	1	—	—
September	3	0	—	—
October	0	−3	−3	9
November	6	3	—	—
December	5	2	—	—
Sum				**84**

Target semideviation $= \sqrt{\frac{84}{(12-1)}} = 2.7634\%.$

2. If the target return were 4 percent, would your answer be different from that for question 1? Without using calculations, explain how would it be different?

   Solution:

   If the target return is higher, then the existing deviations would be larger and there would be several more values in the deviations and squared deviations below the target; so, the target semideviation would be larger.

How does the target downside deviation relate to the sample standard deviation? We illustrate the differences between the target downside deviation and the standard deviation in Example 4, using the data in Example 3.

EXAMPLE 4

Comparing the Target Downside Deviation with the Standard Deviation

1. Given the data in Example 3, calculate the sample standard deviation.

   Solution:

Month	Observation	Deviation from the Mean	Squared Deviation
January	5	2.75	7.5625
February	3	0.75	0.5625
March	−1	−3.25	10.5625
April	−4	−6.25	39.0625
May	4	1.75	3.0625
June	2	−0.25	0.0625
July	0	−2.25	5.0625
August	4	1.75	3.0625